Active case detection using loop-mediated isothermal amplification technology and treatment versus standard of care for malaria in pregnancy in Ethiopia (LAMPREG): a pragmatic randomised diagnostic outcomes trial

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Summary

Background Malaria in pregnancy is a substantial public health risk, particularly in sub-Saharan Africa. Conventional diagnostic tools, such as microscopy and rapid diagnostic tests (RDTs), often miss infections with low-level parasitaemia. We aimed to evaluate the effectiveness of a screen-and-treat strategy using loop-mediated isothermal amplification (LAMP) technology compared with the standard of care to improve maternal and infant health outcomes in rural Ethiopia, where intermittent preventive treatment in pregnancy is not implemented.

Methods We conducted a pragmatic randomised diagnostic outcomes trial across eight health facilities (three hospitals and five health centres) in Ethiopia. Pregnant women aged at least 18 years in their first or second trimester, established via ultrasound dating, were randomly assigned to the LAMP group (in which patients were tested for malaria with LAMP, RDT, and microscopy at every visit, regardless of the presentation of malaria symptoms) or the standard-of-care group (in which patients were tested for malaria with RDT and microscopy only at visits where they were symptomatic for malaria, which is the standard of care in Ethiopia). Randomisation was done by data managers, stratified by gravidity, and done without masking or concealment. Women who tested positive for malaria infection were treated with artemether–lumefantrine. The primary outcome was the proportion of liveborn infants with low birthweight (<2500 g) in both the intention-to-treat (ITT; all randomised participants) and per-protocol (participants for whom birthweight was recorded) populations. We assessed the diagnostic performance (ie, sensitivity, specificity, and positive and negative predictive values) of LAMP, RDT, and microscopy by comparing them to a molecular reference method (laboratory-developed test reference 28). This trial is registered with ClinicalTrials.gov (NCT03754322) and is completed.

Findings 2425 women were enrolled between June 18, 2021, and July 19, 2023 (1570 [64·7%] randomly assigned to the LAMP group and 855 [35·3%] to the standard-of-care group). We did not meet the prespecified sample size of 2583. After imputation for missing data, the ITT analysis included 1570 women with 1521 livebirths in the LAMP group and 855 women with 842 livebirths in the standard-of-care group. Among these livebirths, the proportion of infants with low birthweight at delivery did not differ between the LAMP group (74 [4·9%] of 1521 infants) and the standard-of-care group (39 [4·6%] of 842 infants; odds ratio [OR] 1·24 [95% CI 0·70–2·18]; p=0·65). The per-protocol analysis had similar results (33 [3·4%] of 968 infants in the LAMP group *vs* 18 [3·4%] of 525 infants in the standard-of-care group; OR 0·99 [0·56–1·82]; p>0·99). In our trial populations, LAMP had better sensitivity (94·1% [95% CI 90·0–96·9]) than microscopy (67·2% [60·3–73·6]) and RDT (67·6% [60·8–74·0]). Specificity was 99·0% (95% CI 98·6–99·3) with LAMP, 99·7% (99·5–99·9) with microscopy, and 99·97% (99·8–100) with the RDT. The positive predictive value of LAMP (84·6% [95% CI 79·2–89·0]) was lower than that of both microscopy (93·2% [87·8–96·7]) and RDT (99·3% [96·1–100·0]). The negative predictive value of LAMP (99·7% [99·4–99·8]) was higher than both microscopy (98·1% [97·6–98·5]) and RDT (98·2% [97·7–98·6]). Common adverse events included headache (95 [6·1%] of 1570 participants in the LAMP group and 37 [4·3%] of 855 participants in the standard-of-care group) and preterm delivery (82 [5·2%] of 1570 participants in the LAMP group and 58 [6·8%] of 855 participants in the standard-of-care group). Severe adverse events included admission to hospital (20 [1·3%] of 1570 participants in the LAMP group and four [0·5%] of 855 participants in the standard-of-care group) and life-threatening conditions (16 [1·0%] of 1570 participants in the LAMP group and two [0·2%] of 855 participants in the standard-of-care group). No adverse events were related to the LAMP diagnostic intervention.

Interpretation There was no benefit of LAMP over standard of care regarding low birthweight in the trial. LAMP was more accurate at detecting malaria in pregnancy when compared with microscopy and RDT. Future, powered studies should evaluate the effects of highly sensitive malaria tests, including in other transmission settings.

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Introduction

Malaria remains one of the major public health challenges and a leading cause of morbidity and mortality in low-income and middle-income countries, although it is a preventable and treatable disease.[1] Globally, there were 263 million people infected with malaria in 2023, an increase of 11 million since 2022.[1] Sub-Saharan Africa continues to have the highest burden of malaria worldwide, contributing 95% of the morbidity and 96% of all malaria deaths.[1] Malaria mortality remains a substantial threat to people who are pregnant and infants across sub-Saharan Africa.[1,2] In 2023, 36 million women in sub-Saharan Africa lived in 33 malaria-endemic areas, where approximately 34·4% of women had malaria infection during pregnancy.[1]

In Ethiopia, malaria remains a substantial public health concern, with an overall incidence of 47·2 cases per 1000 population at risk and a mortality rate of 0·42 deaths per 100 000 population at risk in 2022–23.[3] The Gambella region recorded the highest burden of 307·2 cases per 1000 and 3·2 deaths per 100 000 population at risk.[3] Malaria contributes 3.0% of all morbidity in all women and 12·7% among pregnant women across the country.[4,5]

Pregnancy itself is a risk factor for malaria, with people who are primigravid being more susceptible to malaria than those who have multigravida.[4] Malaria in pregnancy has adverse consequences for the fetus, although previous exposure appears to have protective benefits, especially in people who have multigravida. Adverse consequences of malaria in pregnancy include low birthweight, pregnancy loss, preterm delivery, maternal or neonatal mortality, fetal growth restrictions, and neurocognitive developmental challenges. This association was seen in a cross-sectional study conducted in some regions of Ethiopia where both placental and peripheral malaria were associated with premature delivery, low birthweight, stillbirth, and maternal anaemia.[6]

To reduce the adverse events of malaria in pregnancy, WHO recommends the use of insecticide-treated nets, chemoprevention such as monthly spaced doses of intermittent preventive treatment in pregnancy using sulfadoxine–pyrimethamine (IPTp-SP) beginning from second trimester, and intermittent screening and treatment in pregnancy during antenatal care using rapid diagnostic tests (RDTs) and artemisinin-based combination therapy.[7] However, IPTp-SP is increasingly challenged by reports of suboptimal (ie, less than 50%) uptake despite other make-up initiatives, such as community IPTp-SP.[8,9] Emerging resistance, particularly driven by mutations in the *pfdhfr* and *pfdhps* genes, can result in partial clearance of parasites[10] and treatment failure.[11,12] Sulfadoxine–pyrimethamine is not widely effective against *Plasmodium vivax*, which is responsible for nearly 40% of malaria cases in Ethiopia, further complicating malaria-prevention efforts in pregnancy. Additionally, unlike many other malaria-endemic countries in sub-Saharan Africa, Ethiopia does not implement IPTp-SP, instead relying on bednets and passive case detection as primary prevention strategies for malaria in pregnancy.[13] Thus, the absence of a dedicated chemoprevention strategy is a crucial gap in malaria-control

Research in context

Evidence before this study

Trials to evaluate outcomes associated with nucleic acid testing for malaria are rare. Based on a comprehensive literature review, previous studies have suggested that malaria detection through molecular methods could improve pregnancy outcomes by enabling earlier treatment. However, evidence on whether superior diagnostic sensitivity translates into clinically significant improvements in neonatal outcomes, such as birthweight, remains inconclusive. Although some observational studies have linked undetected malaria infections with increased risk of low birthweight, to our knowledge, the effect of enhanced case detection via loop-mediated isothermal amplification (LAMP) on pregnancy outcomes has not been rigorously evaluated in a controlled diagnostic trial.

Added value of this study

To our knowledge, this trial is the largest study to evaluate outcomes associated with malaria in pregnancy based on the diagnostic test used. In our trial, LAMP had better sensitivity in detecting asymptomatic low-level malaria infections than standard of care, but there was no difference in the prevalence of low birthweight between LAMP and standard of care.

Implications of all the available evidence

This trial emphasises the value and challenges of outcomes-based research in resource-limited settings by showing molecular testing is feasible in rural health settings, but trials of this scale are difficult to perform in resource-limited settings. In pregnancy, our data suggest consideration should be given to implementation of sensitive diagnostic tests to detect low-level malaria infections in specific transmission settings.

efforts, emphasising the need for research into alternative tools to protect pregnant women from malaria-related adverse outcomes.

The screen-and-treat approach is particularly effective and important for people who present for their first antenatal care visit in the first trimester,[14] especially in sub-Saharan Africa where an estimated 55–76% of pregnant women use skilled antenatal-care services. Because of the substantial limitations of current diagnostic methods, such as microscopy and RDTs, which often lack sufficient quality assurance and analytical sensitivity, low-level infections and asymptomatic malaria frequently evade detection.[15–17] This limitation is particularly of concern as undetected infections are associated with placental inflammation, fibrosis, and functional insufficiency, and therefore miscarriage, preterm delivery, low birthweight, and peripartum haemorrhage, further showing the importance of improving diagnostic capabilities.[11,18–21] Moreover, asymptomatic infections are crucial for disease burden and are reservoirs for malaria transmission, complicating elimination efforts.[22]

To address these diagnostic limitations, we conducted a pragmatic randomised diagnostic outcomes clinical trial across different malaria transmission settings in rural Ethiopia using loop-mediated isothermal amplification (LAMP) diagnostic technology. LAMP is effective in detecting low-level infections, which often evade the standard-of-care method of RDT and microscopy screening. Our main hypothesis was that active case detection of malaria in pregnant women using molecular assays with LAMP at rural health facilities and hospitals would improve mother and child health outcomes in the absence of intermittent preventive therapy.

This trial aimed to evaluate the effectiveness of a screen-and-treat strategy for malaria in pregnancy using LAMP technology to establish whether active case detection of women with asymptomatic malaria via LAMP would lead to improved outcomes for mothers and infants and to establish whether LAMP would better detect malaria than standard-of-care microscopy and RDTs.

Methods

Trial design and participants

We conducted a pragmatic randomised diagnostic outcomes trial in eight health-care facilities (three hospitals and five health centres) across three malaria-transmission settings in rural Ethiopia. Specific health centres (HCs) were Chisabay, Hamusit, Andasa, Uffa, and Kurgen; specific hospitals were Gambella General, Gambella Primary, and Gebretsadik Shawo General (appendix pp 4, 9). The eight trial sites are in three regional states: Amhara, Gambella, and the South West Ethiopia Peoples' Region (SWEPR). During the trial period, on the basis of evaluation of malaria-incidence logs (clinic data reviewed by study team), the SWEPR sites (Uffa HC and Gebretsadik Shawo Hospital) were considered low-transmission sites, the Amhara sites (Chisabay HC, Hamusit HC, and Andasa HC) were considered moderate-transmission sites, and the Gambella sites (Kurgen HC, Gambella General Hospital, and Gambella Primary Hospital) were considered high-transmission sites (appendix pp 4, 9). For statistical analyses, the low-transmission and moderate-transmission sites were combined into a single low-to-moderate group.

In the low-to-moderate transmission sites of Amhara and SWEPR, where malaria transmission is seasonal, enrolment was seasonal and started on June 18, 2021, aligning with the malaria peak season of Ethiopia (September to December). By contrast, the Gambella region has perennial malaria transmission year-round; therefore, enrolment continued throughout the year. The enrolment period was extended due to political instability and the COVID-19 pandemic, adversely affecting participation and staffing.

We enrolled pregnant women presenting at antenatal care clinics across trial sites. Inclusion criteria were being in the first or second trimester of pregnancy, being aged at least 18 years, having the ability to provide informed consent, and agreement to return to the same antenatal care clinic for follow-up visits, delivery, and newborn check 28 days after delivery. Exclusion criteria were antimalarial medication intake currently or within the past 3 weeks, signs and symptoms of severe malaria, high-risk pregnancy as defined by the Ethiopia Ministry of Health guidelines, ultrasound showing multiple pregnancy, and severe anaemia (ie, haemoglobin of 70 g/L or less). Gestational age was assessed by a trained medical officer using a portable, cloud-connected ultrasound device (Clarius C3, Clarius, Vancouver, BC, Canada) after training by qualified radiologists.

The trial protocol (PO/05/20) was approved by the University of Calgary Conjoint Health Research Ethics Board (REB21–0234), the Armauer Hansen Research Institute/All-Africa Leprosy, Tuberculosis, and Rehabilitation Training Centre (AHRI/ALERT) Ethics Review Committee (AAERC) (FORM AF-10–015.1), the Ethiopia Ministry of Science and Higher Education National Research Ethics Review Committee (SRA/11.7/7115/20), and the Ethiopia Food and Drug Administration (02/25/33/I).

After ethical and regulatory approval, but before trial start, community engagement programmes were conducted at all trial sites to inform the community about the trial and receive their feedback (RFG, BT); patient input was represented by health-centre staff. The trial was conducted according to the Helsinki Declaration, Good Clinical Practice, and Good Laboratory Practice, and all staff were certified accordingly. This trial is registered with ClinicalTrials.gov (NCT03754322) and is completed.

Participants received oral and written information regarding the trial in Amharic, after which they had to sign the consent form before conducting any trial-related activity. For participants with illiteracy, a family member observed the informed consent procedure and obtained consent using the participant's fingerprint. Deviations from the trial protocol[23] are included in appendix (p 2).

See Online for appendix

Randomisation and masking

Pregnant women were randomly assigned to the LAMP intervention (in which patients were tested for malaria with LAMP, RDT, and microscopy at every visit, regardless of the presentation of malaria symptoms, as defined in the next section) or to standard of care alone (in which patients were tested with RDT and microscopy at symptomatic visits only, which is the standard of care in Ethiopia). Randomisation was done via a defined sequence implemented in the REDCap electronic data capturing system (version 8.11.5) and was stratified by pregnancy (ie, primigravid *vs* multigravid) to ensure balanced distribution across trial groups, considering increased risk of placental malaria in first pregnancies because of the absence of pregnancy-specific immunity. A data manager (HT) centrally managed randomisation at the coordinating centre (AHRI) for all sites, then provided results to the respective trial sites. Enrolment was done locally at each health facility by trial staff assisted by midwives or health officers at the respective sites. We generated a random sequence for all participants, dividing them into two main groups (ie, LAMP or standard of care). Then, within each group, we stratified by gravidity (ie, first, second, third, fourth, or more than fourth pregnancy). Finally, we randomly assigned participants within each stage of pregnancy. No masking or concealment was done.

Procedures

Women were enrolled at the first antenatal care visit and followed up for up to four planned visits, depending on the timing of their initial enrolment. The visit schedule was: visit 1 at 0–13 weeks and 6 days, visit 2 at 14–27 weeks and 6 days, visit 3 at 28–35 weeks and 6 days, and visit 4 at 36–38 weeks and then at delivery day, with a final follow-up visit at 28 days postpartum. Participants were considered to have completed the trial if birthweight was recorded. We used standardised paper-based and electronic (ie, on Samsung tablet devices) case report forms designed in REDCap to collect sociodemographic characteristics and clinical data through medical history and physical examination conducted by a medical officer or nurse. Sex or gender data were self-reported by study participants. Data on maternal ethnicity were not collected, as it was considered not relevant to this study. Key clinical data collected included age, gravidity, and gestational age at enrolment via ultrasound guidance. Additional laboratory data were haemoglobin and anaemia status. Other demographic information related to participants' residence and contact information were obtained to enhance trial adherence and follow-up. Women were compensated for travel costs, also to improve adherence.

We recorded participants' height and weight at each visit using standard scales available at the health centres. Newborn weight was recorded during the head-to-toe assessment at delivery before first feeding. Digital scales (Heuer, Model iR721B-21, Brussels, Belgium), calibrated to three decimal places, were implemented in the health facilities to ensure standardised and uniform measurement of newborn weight. Haemoglobin levels were measured at each antenatal care visit for women in both groups using a point-of-care method (HemoCue Hb 201+ Analyzer, HemoCue, Ängelholm, Sweden). Both adverse and severe adverse events were collected at each in-person visit based on prespecified criteria and reported per protocol.

In the standard-of-care group, women were tested for *Plasmodium* infection with microscopy and RDT only when they presented at antenatal-care visits with at least one of the following malaria symptoms: fever, chills, rigours, sweating, severe headache, generalised body and joint pain, or pallor, following the malaria diagnosis and treatment guideline from the Ethiopia Federal Ministry of Health. By contrast, the LAMP group used a CE-approved LAMP malaria test at every antenatal care visit, regardless of symptoms, in addition to RDT and microscopy. All commercial tests were performed according to the manufacturer's instructions.

To run these malaria tests, trained professionals collected whole-blood samples at each visit. The RDT used was the CareStart Malaria HRP2/pLDH (Pf/Pv) COMBO (Access Bio, Somerset, NJ, USA) to distinguish between *Plasmodium falciparum* and *P vivax* malaria. To obtain microscopy results, both thick and thin blood smears were prepared and appropriately labelled. After air-drying, thin films were fixed with absolute methanol for 5 sec while both thin and thick blood films were stained with freshly prepared 10% Giemsa working solution diluted with buffered water (pH 7·1–7·2) for 10 min and examined under 100× magnification. Thin films were used to identify *Plasmodium* parasite species, whereas the thick films were examined to detect and measure the density of the parasite against 200 white blood cells, assuming the mean white blood cell count was 8000 cells per µL (parasites per µL=number of asexual parasites × 8000/number of white blood cells counted) as per WHO recommendations (WHO, 2015).[24] The slides were read by trained laboratory technicians under a light microscope (Olympus CX23, Bartlett, TN, USA), and positive results for malaria parasites were recorded, indicating parasite density (counts) against white blood cells on thick film. Slides were reported as negative after examining 8000 white blood cells. For malaria microscopic examination, slides were read by two trained laboratory technologists. For discrepant results between two readers, a third reading done by an independent WHO-certified expert at Amhara Public Health Institute and Jimma University was considered final.

The LAMP test consisted of up to three different assays run by laboratory technologists on the HumaLoop M instrument (HUMAN Diagnostics Worldwide, Wiesbaden, Germany). First, the Loopamp Malaria Pan Detection Kit (Eiken Chemical, Tokyo, Japan) was used to detect the presence of malaria generally (it yields a positive result if any of *P falciparum, P vivax, Plasmodium ovale, Plasmodium knowlesi,* or *Plasmodium malariae* are present). If this test was positive, then the Loopamp Malaria Pf Detection Kit and the Loopamp Malaria Pv Detection Kit were used to detect

the presence of *P falciparum* and *P vivax*, respectively. For quality assurance, each test run included positive and negative controls. Additionally, 10% of all negative peripheral blood samples underwent double reading at a reference centre at the regional reference laboratory in Bahir Dar and Jimma.

Discordant results among LAMP, RDT, and microscopy were resolved with a previously validated[25] highly sensitive molecular method, which served as the reference. This molecular method is an independent ultra-sensitive loop-mediated isothermal amplification laboratory-developed test. After the trial had concluded, blood samples from the standard-of-care group that were negative for malaria were tested with this same molecular method to understand the degree to which asymptomatic malaria was missed in this group. At delivery, placental blood was collected by trained technologists and also tested for malaria using this same molecular test.

Outcomes

The primary outcome of the trial was the proportion of live newborns with low birthweight (<2500 g). Stillbirths and miscarriages were not included in the denominator of this proportion. Secondary outcomes were birthweight; maternal haemoglobin concentrations during pregnancy and at delivery; infant haemoglobin concentrations at delivery; pregnancy loss (ie, stillbirths, miscarriages, and abortions); neonatal death (excluding stillbirths and miscarriages); premature birth (defined as gestational age at birth <37 weeks); small for gestational age (defined as having birthweight below the 10th percentile at the given gestational age, according to data from the Intergrowth website, which were not specific to Ethiopian babies); and signs of *Plasmodium* infection, which we report as the presence of placental malaria. All secondary outcomes, like the primary outcome, were computed using only livebirths, except for the outcome of pregnancy loss, for which the denominator was set to the total number of women who experienced livebirths or pregnancy loss (in each group). Because most participants did not attend all antenatal care visits, the maternal haemoglobin concentration during pregnancy for each participant was computed by averaging the maternal haemoglobin concentration across all their antenatal care visits before the birth visit. Sensitivity, specificity, positive predictive value, and negative predictive value were evaluated for LAMP, RDT, and microscopy by comparison with a molecular reference method.[25] Adverse events and severe adverse events were managed by the medical officer and reported to study leads and principal investigators for further action.

Statistical analysis

We calculated the sample size to detect a clinically significant difference in the proportion of newborns with low birthweight between the LAMP and standard-of-care groups. Assuming a baseline prevalence of low birthweight of 17·3% in Ethiopia,[15] we targeted an absolute difference of 5% with a significance level of 0·05. Anticipating a 20% attrition rate, we established that a total sample size of 2583 participants would be necessary to achieve 80% statistical power.

We used R version 4.2.2 for all statistical analyses. Categorical variables were summarised as n (%), with comparisons made using the χ^2 test. For categorical primary and secondary outcome variables, we also reported odds ratios (ORs). ORs were calculated via a generalised linear model with a logit link function, with trial group as a fixed effect. 95% CIs were calculated from the Wald test. Continuous variables were summarised as mean (SD) and were compared via a *t* test. p values were computed via Student's *t* test. For n (%) data, p values were calculated via χ^2 test.

All primary and all secondary outcomes (except for placental malaria, for reasons described below) were investigated in both the intention-to-treat (ITT) and per-protocol populations, whereas post-hoc outcomes were only analysed per protocol. Adverse and severe adverse events were monitored and reported as stated in the protocol. The ITT population was defined as all participants randomly assigned to each group. For this population, missing data were imputed using multiple imputation with the mice package in R. The per-protocol population was defined as all participants for whom a birthweight was recorded, and missing data were not imputed. The main outcomes affected by the imputation were birthweight, gestational age at birth, pregnancy loss, neonatal death, and maternal and newborn haemoglobin concentrations at birth. Because placental malaria outcomes were only recorded in 1239 (51·1%) of 2425 samples, and, among those that were recorded, there were only nine (0·7%) positives of 1239 samples, we did not think the imputation model would accurately impute new values for this variable. Therefore, we only performed a complete case analysis for placental malaria in the per-protocol population and did not include it in the ITT analysis. Here, complete cases are defined as those for which the result of the placental malaria test was recorded. To further understand potential differences in outcomes between women who completed the study (ie, who had a recorded birthweight measurement) and those who did not, univariate analyses of several variables correlated with primary and secondary outcomes were conducted between the two groups (appendix p 4). These variables (which were prespecified) included the proportion in the standard-of-care group, age at enrolment, material haemoglobin and anaemia at enrolment, the proportion of multigravid women, the gestational age at enrolment, and the transmission setting. Anaemia was defined as a haemoglobin concentration less than 11.0 g/dL.

For the primary analysis of the proportion of newborn babies with low birthweight, additional adjustments were made. We observed that birthweights were frequently rounded to the nearest 100 g and occasionally rounded to the nearest 500 g when recorded. These procedures differed across sites and were inferred from visual inspection

For the **Intergrowth website** see https://intergrowth21.com/tools-resources/newborn-size

of data distributions and clustering at specific values (2500 g, 3000 g, etc). Because of the frequent rounding of birthweight measurements to these values, we attempted to more accurately capture the underlying birthweight distribution using a finite mixture model with two normal distributions. This two-component normal distribution was used because we assumed the dataset was composed of a normally distributed primary population in which most newborns were included, and a second, normally distributed low-birthweight population with mean weight of less than 2500 g.[26] Following the fit of the finite mixture model to the LAMP and standard-of-care birthweight distributions separately, we obtained a point estimate of the proportion of low birthweight in each group as the value of the cumulative distribution function at 2500 g (appendix p 3). These proportions were then converted to rounded estimates of the low-birthweight count in each group by multiplying them by the respective sample sizes of the two groups. The resultant counts of the two groups were then compared with a χ^2 test, and an OR was calculated.

We assessed the sensitivity, specificity, and positive and negative predictive values of LAMP, RDT, and microscopy by comparing them to ultra-sensitive molecular testing using the molecular reference method. Only LAMP group data were used for these computations because the standard-of-care group was not tested with LAMP. Diagnostic performance data were reported with Clopper–Pearson 95% CIs.

Diagnostic data were further analysed in three sets of prespecified subgroups. First, women who tested positive for malaria were broken down into two groups: those who were symptomatic for at least one visit and those who were asymptomatic at every visit. Here, symptomatic was defined as displaying any of the following signs or symptoms at the time of the positive test: fever, chills, rigours, sweating, severe headache, generalised body and joint pain, or pallor. Individual visits that were positive for malaria were classified as symptomatic if the participant had at least one of the above malaria symptoms at the given visit, and asymptomatic if they had none. The number of positive or negative tests using each of the three diagnostic methods (RDT, microscopy, and LAMP) were tallied for each group. Second, individual tests were broken down into those positive for *P falciparum*, those positive for *P vivax*, and those positive for both (mixed infections). The number of such tests in each group was stratified separately by health-care centre and antenatal care visit and tallied. Third, women who tested positive for malaria were broken down into those who only tested positive with LAMP and those who tested positive with RDT or microscopy (regardless of whether or not they tested positive with LAMP). The symptom status of these two groups was tallied and compared. Furthermore, primary and secondary outcomes were compared between those who tested positive for malaria with LAMP in the LAMP group and those who only tested positive for malaria in the standard-of-care group with the reference method. The latter were referred to as undiagnosed since they did not test positive for malaria with RDT or microscopy during the trial and were therefore never treated for malaria. Between these groups, the following variables were compared using a *t* test or χ^2 test where applicable: newborn haemoglobin at birth, newborn anaemia at birth, birthweight, low birthweight proportion, newborn haemoglobin 28 days after birth, weight gain from birth to 28 days after birth, pregnancy loss (miscarriage, stillbirth, or abortion), placental malaria and preterm delivery. Only complete cases (ie, those for which the given variable was recorded) were included for this analysis. Post hoc, we further disaggregated the diagnostic performance by transmission setting (low-to-moderate *vs* high) to establish the eco-epidemiological efficacy of each test. Sensitivity, specificity, and positive and negative predictive values were computed separately for these two transmission settings. We also intended to analyse diagnostic results in women who gave birth to babies with low birthweight, but there was an insufficient number of these women in the study for such an analysis (33 [2·1%] of 1570 participants in the LAMP group and 18 [2·1%] of 855 participants in the standard-of-care group).

Post hoc, the tRNA ligase concentration of each positive malaria sample was determined via droplet digital PCR. Concentration differences were evaluated with the Mann–Whitney test between the following three pairs of groups: (1) symptomatic and asymptomatic individuals; (2) individuals who tested positive for malaria with at least one diagnostic test during the trial and those who only tested positive with the reference method, and (3) individuals in low-to-moderate and high transmission settings.

Interim analyses were conducted at 25% of the inclusion target and completion target by the study team using the aforementioned analysis plan.

Infant weight gain from birth to day 28 was examined post hoc as an exploratory outcome with multivariable linear regression models adjusted for gravidity, maternal anaemia status, transmission setting, maternal age, number of visits before delivery, and symptom status. At birth, neonatal weight can vary due to multiple factors, including breastfeeding initiation, but by 28 days, weight gain stabilises, potentially revealing the impact of low-level malaria infection. We also fit adjusted multivariable linear regression models stratified according to transmission setting (ie, low-to-moderate *vs* high). Variables were also considered as potential modifiers using interaction terms. Post-hoc outcomes were only analysed in the per-protocol population, so only included infants for whom a recorded birthweight was present.

Data entered into REDCap were cross-verified against the original source documents by data management staff (HT) at the trial sites. We held weekly data quality-control meetings with the trial team throughout the trial to address and resolve queries and discrepancies. Final statistical analyses were conducted on the verified data on REDCap after data cutoff on July 19, 2023.

Role of the funding source
The funders of the trial had no role in trial design, data collection, data analysis, data interpretation, or writing of the report.

Results
The number of women who were screened and decided not to enrol were not captured. Between June 18, 2021, and July 19, 2023, 2425 pregnant women were enrolled in the trial (figure 1). Of these 2425 participants, 1570 (64·7%) were randomly assigned to the LAMP group and 855 (35·3%) to the standard-of-care group. Baseline sociodemographic characteristics were similar between the two groups (table 1). Mean maternal age was 24·6 years (SD 4·5) in the LAMP group and 24·7 years (4·6) in the standard-of-care group, and 1612 (66·5%) of 2425 participants were multigravida. Maternal haemoglobin concentrations were 12·3 g/dL (1·4) in both groups and gestational age at enrolment was similar between the two groups (18·0 weeks [5·1] in the LAMP group and 18·0 [5·3] in the standard-of-care group; table 1). Treatment was mostly with artemether–lumefantrine, as per national guidelines (appendix p 4). The number of participants who completed the trial was 968 [61·7%] of 1570 in the LAMP group and 525 [61·4%] of 855 in the standard-of-care group (figure 1). There were significant differences in enrolment criteria between those who did and those who did not complete the study, specifically with respect to anaemia rate, haemoglobin concentration, and transmission setting (appendix p 4). As of the data cutoff (July 19, 2023), median follow-up, defined as median time from screening visit to final visit, was 21·4 weeks (IQR 14·9–26·9) for the LAMP group and 20·9 weeks (IQR 13·6–26·0) for the standard-of-care group.

After imputation for missing data, the ITT analysis included 1570 women with 1521 livebirths in the LAMP group and 855 women with 842 livebirths in the standard-of-care group. Among these livebirths, the crude proportion of infants with low birthweight at delivery did not differ between the LAMP group (74 [4·9%] of 1521 infants) and the standard-of-care group (39 [4·6%] of 842 infants; OR 1·24 [95% CI 0·70–2·18]; p=0·65; table 2). The adjusted proportion of infants with low birthweight at delivery did not differ between the LAMP group (104 [6·8%] of 1521 infants) and the standard-of-care group (55 [6·5%] of 842 infants; OR 1·28 [0·79–2·09]; p=0·67; table 2). The per-protocol analysis was similar (50 [5·2%] of 968 infants in the LAMP group *vs* 27 [5·1%] of 525 infants in the standard-of-care group; OR 1·00 [0·63–1·64]; p>0·99). The crude proportion of infants with low birthweight also did not differ by group in the per-protocol analysis (33 [3·4%] of 968 infants in the LAMP group *vs* 18 [3·4%] of 525 infants in the standard-of-care group; OR 0·99 [0·56–1·82]; p>0·99; table 2). Mean birthweight on day 0 was not different between groups (3191 g [SD 614] in the LAMP group *vs* 3179 g [580] in the standard-of-care group; p=0·62; table 2). The three most frequently recorded birthweights, 3000 g, 3200 g, and 3500 g, comprised 596 (39·9%) of the 1493 recorded birthweights.

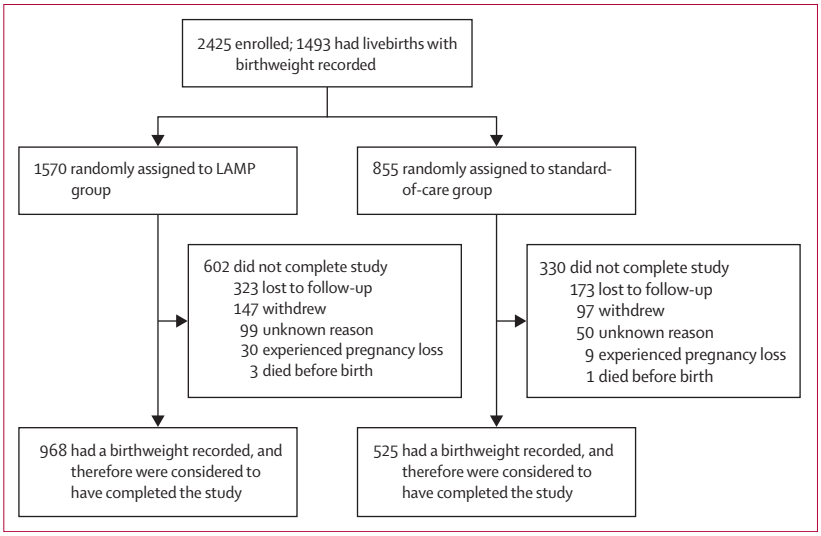


Figure 1: **Trial profile**
Participants were considered to have completed the study if a birthweight measurement was recorded. LAMP=loop-mediated isothermal amplification.

	LAMP group (n=1570)	Standard-of-care group (n=855)
Maternal age, years	24·6 (4·5)	24·7 (4·6)
Maternal haemoglobin, g/dL	12·3 (1·4)	12·3 (1·4)
Maternal anaemia	241 (15·4%)	115 (13·5%)
Multigravida	1036 (66·0%)	576 (67·4%)
Gestational age, weeks	18·0 (5·1)	18·0 (5·3)
First trimester	228 (14·5%)	134 (15·7%)
Second trimester	1342 (85·5%)	721 (84·3%)
Symptomatic for malaria at screening	66 (4·2%)	37 (4·3%)
Member of household positive for malaria in past month	15 (1·0%)	9 (1·1%)
Previous pregnancy complications (at least one previous instance of miscarriage, stillbirth, or neonatal death)	187 (11·9%)	114 (13·3%)
Included in per-protocol analysis	968 (61·7%)	525 (61·4%)
Number of antenatal visits before delivery		
0	1 (0·1%)	0
1	361 (23·0%)	211 (24·7%)
2	446 (28·4%)	238 (27·8%)
3	588 (37·5%)	326 (38·1%)
4	171 (10·9%)	80 (9·4%)
5	3 (0·2%)	0

Data are mean (SD) or n (%). Data on maternal ethnicity were not collected. LAMP=loop-mediated isothermal amplification.

Table 1: **Baseline demographic and clinical data of participants in the intention-to-treat population**

None of the secondary outcomes were different between the LAMP and standard-of-care groups (table 2).

Common adverse events included headache (95 [6·1%] of 1570 participants in the LAMP group and 37 [4·3%] of 855 participants in the standard-of-care group) and preterm delivery (82 [5·2%] of 1570 participants in the LAMP group and 58 [6·8%] of 855 participants in the standard-of-care group). Severe adverse events included admission to hospital (20 [1·3%] of 1570 participants in the LAMP group and four [0·5%] of 855 participants in the standard-of-care

	LAMP group	Standard-of-care group	p value	OR (95% CI)
Intention-to-treat analysis				
Primary outcomes				
Crude low birthweight (ie, <2500 g)	74/1521 (4·9%)	39/842 (4·6%)	0·65	1·24 (0·70–2·18)
Adjusted low birthweight (ie, <2500 g)	104/1521 (6·8%)	55/842 (6·5%)	0·67	1·28 (0·79–2·09)
Secondary outcomes				
Newborn birthweight, g	3191 (614)	3179 (580)	0·62	NA
Maternal haemoglobin at birth, g/dL	12·0 (2·2)	12·1 (2·3)	0·15	NA
Newborn haemoglobin at birth, g/dL	18·1 (3·6)	18·1 (3·9)	0·67	NA
Pregnancy loss (ie, miscarriage, stillbirth, or abortion)	49/1570 (3·1%)	13/855 (1·5%)	0·051	1·85 (0·88–3·90)
Neonatal death	10/1521 (0·7%)	5/842 (0·6%)	0·78	1·33 (0·26–6·88)
Preterm birth (ie, <37 weeks)	146/1521 (9·6%)	97/842 (11·5%)	0·25	0·76 (0·55–1·06)
Small for gestational age	276/1521 (18·1%)	149/842 (17·7%)	0·60	1·07 (0·80–1·43)
Per-protocol analysis				
Primary outcomes				
Crude low birthweight (ie, <2500 g)	33/968 (3·4%)	18/525 (3·4%)	>0·99	0·99 (0·56–1·82)
Adjusted low birthweight (ie, <2500 g)	50/968 (5·2%)	27/525 (5·1%)	>0·99	1·00 (0·63–1·64)
Secondary outcomes				
Newborn birthweight, g	3230 (446)	3200 (420)	0·21	NA
Maternal haemoglobin at birth, g/dL	12·1 (1·5)	12·2 (1·5)	0·38	NA
Newborn haemoglobin at birth, g/dL	18·3 (2·4)	18·2 (2·6)	0·79	NA
Pregnancy loss (ie, miscarriage, stillbirth, or abortion)	30/998 (3·0%)	9/534 (1·7%)	0·16	1·81 (0·89–4·07)
Neonatal death	5/968 (0·5%)	2/525 (0·4%)	>0·99	1·36 (0·29–9·50)
Preterm birth (ie, <37 weeks)	82/968 (8·5%)	58/525 (11·0%)	0·12	0·75 (0·52–1·07)
Small for gestational age	154/968 (15·9%)	81/525 (15·4%)	0·87	1·04 (0·78–1·39)
Placental malaria*	7/828 (0·8%)	2/411 (0·5%)	0·73	1·74 (0·42–11·74)

Data are mean (SD) or n/N (%), in which N is either number of livebirths or number of women, depending on the analysis, unless otherwise indicated. Crude low-birthweight frequency used recorded birthweights, whereas adjusted low-birthweight frequency used estimated low-birthweight proportion computed after removal of heaping by fitting a two-component normal mixture model to the birthweight distribution of each group. The per-protocol analysis was conducted using only participants for whom birthweight was recorded. LAMP=loop-mediated isothermal amplification. NA=not assessed. OR=odds ratio. *Placental malaria was not analysed in the intention-to-treat population and was analysed via complete case analysis in the per-protocol population.

Table 2: **Outcomes**

group) and life-threatening conditions (16 [1·0%] of 1570 participants in the LAMP group and two [0·2%] of 855 participants in the standard-of-care group). No adverse events were related to the LAMP diagnostic intervention (appendix p 8). No dose reductions, discontinuations for drug-related toxic effects, or treatment-related deaths were observed (data not shown).

Diagnostic performance analyses showed that LAMP had better sensitivity (192/204, 94·1% [95% CI 90·0–96·9]) than microscopy (137/204, 67·2% [60·3–73·6]) and RDTs (138/204, 67·6% [60·8–74·0]) across all three transmission settings. All three methods had a specificity of 99% or higher (LAMP 3471/3506, 99·0% [95% CI 98·6–99·3]; microscopy 3496/3506, 99·7% [99·5–99·9]; RDT 3505/3506, 99·97% [99·8–100·0]; figure 2B; appendix p 5). The positive predictive value of LAMP (192/227, 84·6% [95% CI 79·2–89·0]) was lower than that of both microscopy (137/147, 93·2% [87·8–96·7]) and RDTs (138/139, 99·3% [96·1–100·0]). The negative predictive value of LAMP (3471/3483, 99·7% [95% CI 99·4–99·8]) was higher than both microscopy (3496/3563, 98·1% [97·6–98·5]) and RDT (3505/3571, 98·2% [97·7–98·6]). In both low-to-moderate and high-transmission settings, LAMP showed consistent sensitivity, whereas microscopy and RDTs were less sensitive in low-to-moderate settings than in high-transmission settings (appendix p 5).

The following diagnostic results are all with respect to visits that were positive for malaria after discrepant resolution (ie, true positives). A significantly higher proportion of asymptomatic visits that were positive for malaria occurred in low-to-moderate transmission areas relative to high-transmission areas (figure 2C). Of those visits positive for malaria in the LAMP group, 40 (83·3%) of 48 that were positive by LAMP, but negative by RDT and microscopy were asymptomatic, while only 73 (46·8%) of 156 that were positive by RDT or microscopy were asymptomatic (appendix p 5). Of 113 asymptomatic visits in the LAMP group that were positive for malaria, 60 (53·1%) were detected with microscopy and 57 (50·4%) were detected with RDT, whereas 101 (89·4%) were detected with LAMP (figure 2D; appendix p 5). Most positive malaria cases occurred during visits 2–4 at the antenatal clinic, coinciding with the second and third trimester (appendix p 6). Most malaria episodes were single episodes rather than recurrent in both the LAMP group and standard-of-care group (appendix p 6). In the LAMP group, *P falciparum* (133 [65·2%] of 204 positive tests) was more common than

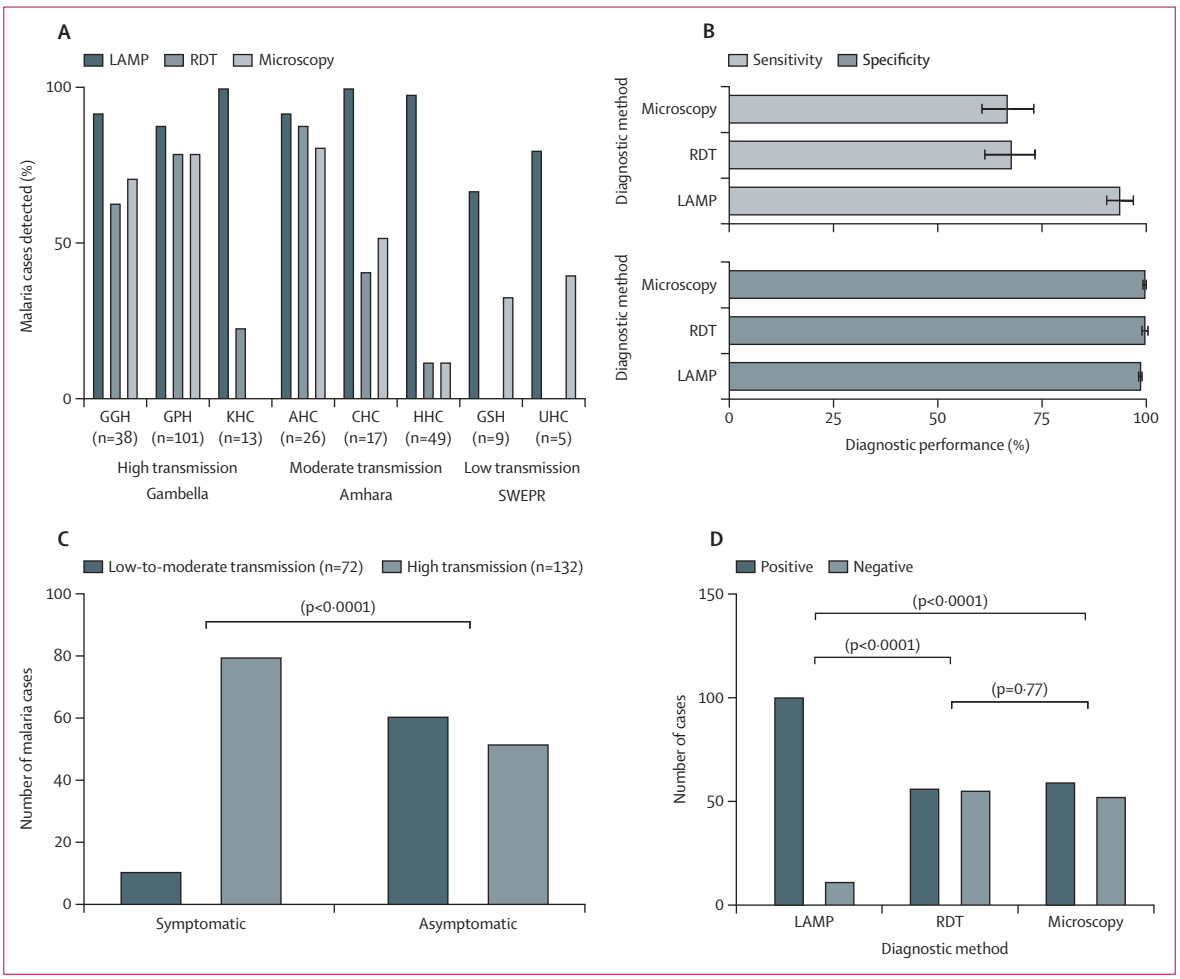


Figure 2: **Diagnostic performance of LAMP, microscopy, and RDT**

(A) Malaria cases (instances of malaria infection) detected at each health centre or hospital by transmission intensity before discrepant resolution. (B) Sensitivity and specificity of each diagnostic method after discrepant resolution with a reference test. Error bars are Clopper–Pearson 95% CIs. (C) Relative detection of malaria cases by transmission setting for symptomatic and asymptomatic malaria after discrepant resolution. (D) Number of asymptomatic malaria cases detected by each method after discrepant resolution. Here, positive and negative refer to whether the given test was positive or negative for malaria. AHC=Andasa Health Centre. CHC=Chisabay Health Centre. GGH=Gambella General Hospital. GPH=Gambella Primary Hospital. GSH=Gebretsadik Shawo Hospital. HHC=Hamusit Health Centre. KHC=Kurgen Health Centre. LAMP=loop-mediated isothermal amplification. RDT=Rapid diagnostic test. SWEPR=South West Ethiopia Peoples' Region. UHC=Uffa Healthcare Centre.

P vivax (40 [19·6%] tests) and mixed infections (23 [11·3%] tests). Eight of the 204 true positives could not be resolved as *P falciparum*, *P vivax*, or mixed. Moreover, in the standard-of-care group, *P falciparum* (21 [70·0%] of 30 positive tests) was more common than *P vivax* (two [6·7%] tests) and mixed infections (four [13·3%] tests). Three of the 30 true positives could not be resolved as *P falciparum*, *P vivax*, or mixed. For missed diagnoses in the standard-of-care group, 76 (66·7%) of 114 tests showed *P falciparum*, which was more common than *P vivax* (31 [27·2%]) and mixed infections (seven [6·1%]; appendix p 6). 51 (67·1%) of 76 missed *P falciparum* diagnoses and 20 (64·5%) of 31 missed *P vivax* diagnoses occurred in the northwest Ethiopian sites of Andhasa, Chisabay, and Hamusit (appendix p 7).

Post-hoc analysis of participants with false negative RDT results showed that most were false negative due to low-level infections and a minority were due to *hrp2* deletions (appendix p 11). Parasite concentrations were on average lower in women undiagnosed in the standard-of-care group (ie, those who tested positive for malaria with the reference method, but not RDT or microscopy) than in those who tested positive for malaria by any diagnostic method during the trial (appendix p 12). Moreover, parasite concentrations were on average lower in low-to-moderate transmission settings (p<0·0001) and in asymptomatic individuals (p<0·0001) than in high-transmission settings and symptomatic individuals (appendix pp 13–14). Post-hoc exploratory analysis of outcomes in participants who were undiagnosed for malaria in the standard-of-care group but positive using the reference method versus those who tested positive for malaria with LAMP in the LAMP group are shown in the appendix (p 7).

Differences in weight gain from day 0 to day 28 between the LAMP group and standard-of-care group were explored

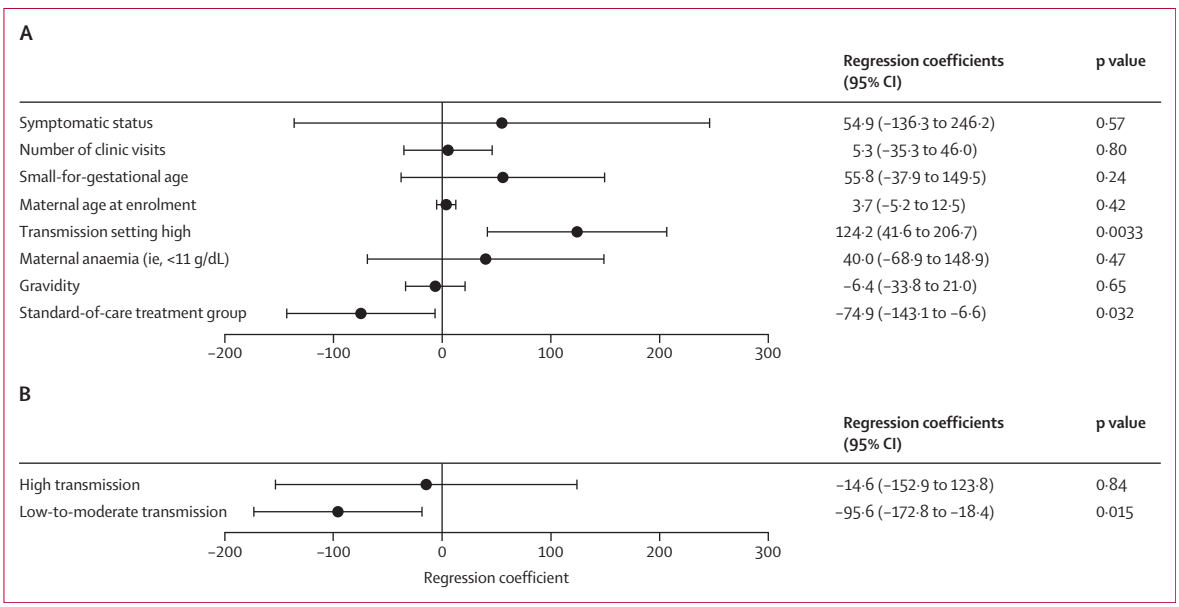


Figure 3: **Post-hoc birthweight and weight gain outcomes by diagnostic method and transmission setting**
(A) Multivariable linear regression of change score in infant weight gain between birth and day 28. Overall p value was 0·0059. (B) Stratified linear regression to distinguish between low-to-moderate transmission and high-transmission settings for weight gain. Overall p value was 0·046 for low-to-moderate transmission and 0·98 for high transmission.

post hoc in infants. 1130 infants had measurements for all relevant variables and were included in the model (728 [64·4%] of 1130 were from the LAMP group and 402 [35·6%] of 1130 were from the standard-of-care group). Infants in the LAMP group gained more weight (mean gain 1061 g [SD 579]) than those in the standard-of-care group (975 g [513]; p=0·013). This difference in weight gain between the two groups persisted after accounting for additional sources of variance with a multivariable linear regression model (figure 3). For infants born appropriate for gestational age (birthweight between 10th and 90th percentiles) but who tested negative for malaria in each group, more weight gain was observed in the LAMP group (mean weight 1147 g [625]) than in the standard-of-care group (926 g [490]; p=0·00011).

Discussion

This trial contributes to the growing body of literature on the importance of improving diagnostic tools for malaria in pregnancy. We found no difference in our primary outcome of occurrence of low birthweight in mothers assigned to LAMP versus standard of care. Additionally, we found no difference between the groups in any of our secondary outcomes. Notably, our study was underpowered, and so our findings should be interpreted with caution.

Our data suggest that LAMP outperformed microscopy and RDTs in some diagnostic performance measures when attempting to detect low-level and asymptomatic malaria infections in pregnant women. However, this trial was underpowered and further research is needed to support our results. Our data support previous research that shows that traditional diagnostic methods often lack sensitivity, particularly in people who are asymptomatic.[15–17,22,25] For example, several studies have found that RDTs and microscopy did not detect a substantial proportion of low-level infections, which can still contribute to maternal anaemia and poor birth outcomes.[16–18] Low-level infections and *hrp2* deleted strains mostly occurred in low-to-moderate transmission settings in this trial and comprised a substantial number of missed diagnoses in the standard-of-care group. Consistent with our findings, a meta-analysis of LAMP studies in pregnancy confirmed that LAMP is more sensitive than microscopy and RDTs.[17] The authors also suggest that large prospective trials are required to confirm the impact of treating low-level infections which our study aimed to do. As part of our post-hoc analyses, weight gain observed among infants in the LAMP group by day 28 postpartum compared with the standard-of-care group could suggest that early detection and treatment of malaria through active case detection with LAMP technology could positively affect infant growth, especially in low-to-moderate transmission settings.[27] This post-hoc finding aligns with similar studies that have shown early and accurate detection of malaria during pregnancy preventing or reducing the adverse effects of malaria on fetal development, including low birthweight and preterm delivery.[19,28]

This finding potentially aligns with the observation of a 2010 Gambian study that showed the long-term effects of placental malaria on infant weight development.[21] The fact that we observed no difference in maternal haemoglobin concentrations between the two groups could be because the effect of LAMP-based detection might be more pronounced in fetal outcomes than in maternal health. The diagnostic performance of LAMP in this trial provides some support

for integrating it into routine antenatal care programmes in low-to-moderate transmission areas where *hrp2* deletions and low-level infections are disproportionately present.

This trial had several limitations. First, placental malaria was evaluated by PCR instead of histology, which might not reflect true placental disease and could explain its low rate of detection in our trial. Second, the proportion of women who were screened and decided not to enrol were not captured, so we could not evaluate bias. Third, only 1493 (61·6%) of 2425 participants had a recorded birthweight and therefore completed the study. This attrition is important and could introduce bias. For example, individuals who did not complete the study had lower maternal haemoglobin concentrations, higher maternal anaemia, and originated disproportionately from high-transmission areas, which could influence the generalisability of the findings. The ITT analysis attempted to account for this bias by imputing missing data. We saw slightly increased rates of low birthweight, small for gestational age, and preterm delivery in the ITT analysis relative to the per-protocol analysis, as well as lower birthweights, all of which are consistent with the direction of this bias. The attrition is not surprising, considering the health sector transformation plan report by the Ethiopian Ministry of Health indicated that antenatal care contacts, skilled birth attendance, and postnatal-care coverage were 43%, 50%, and 34%, respectively.[29] Fourth, because we did not reach our desired sample size of 2583 and our malaria incidence was relatively low overall, this trial was underpowered. The sample size calculation was based on a higher low birthweight incidence than was observed in the trial. This discrepancy could be partly explained by the fact that recorded birthweights were frequently rounded to the nearest 100 g or occasionally to the nearest 500 g. The approximately 2% increase in low birthweight proportion when we used the finite mixture model compared with the recorded birthweights indicates a plausible rate of low birthweight misclassification due to digit preference, although the actual misclassification rate might be higher. Although there was no difference in rounding between the two groups, rounding to the nearest 500 g occurred nearly twice as often in Gebretsadik Shewa Hospital and Uffa Healthcare Centre, both of which were in low-transmission areas (data not shown). This difference could have affected our regression models, which were stratified by transmission status. Nonetheless, because the two groups had similar low birthweight proportions to each other both before and after accounting for heaping with the finite mixture model, we do not believe the primary outcome was affected by digit preference. Rather, the misclassification of low birthweight, particularly due to heaping at 2500 g, appears to have affected both groups relatively equally, resulting in an underestimate of the low birthweight proportion overall. Fifth, external factors such as civil unrest and the COVID-19 pandemic affected participant retention and the overall timeline of the trial, despite extending the timeline beyond schedule. These factors could have affected the consistency of data collection and the ability to maintain strict trial protocols, potentially influencing the results. Sixth, the randomisation and enrolment design differed in practice between sites due to extenuating circumstances: political conflicts resulting in the temporary closure of sites, no women with malaria infection at sites, supply chain challenges for testing reagents, and staffing interruptions to support recruitment. As a result, the trial is pragmatic in nature and reflects the resource-limited setting in which the study was conducted. Seventh, we discussed the effect of conducting our trial across three different transmission settings (ie, low, moderate, and high). Although this provided insight into the effectiveness of LAMP in different settings, it also introduced variability that could have obscured the detection of differences between the LAMP and standard-of-care groups. Finally, overlapping CIs suggested that some findings will not be statistically robust across all settings, as the prevalence of submicroscopic infections varies by region.

Implementation of LAMP in antenatal care in Ethiopia is hampered by cost considerations. RDTs and microscopy are substantially cheaper in terms of consumables than LAMP. However, given the suggested improved detection of low-level infections and the potential effects on neonatal weight gain seen in our post-hoc analyses, further evaluation is required at the research and policy levels.

LAMP detection of malaria in pregnancy did not yield a significant difference in low birthweight frequency compared with standard of care, the primary outcome of the study. Moreover, there was no difference in secondary outcomes. Post-hoc exploratory analyses indicated that LAMP detected more overall malaria, especially low-level malaria infections in low-to-moderate transmission settings, but these should be interpreted with caution. This increased detection could potentially be associated with improved weight gain in the first 28 days after delivery. Further studies should explore these findings in diverse eco-epidemiological settings, especially where IPTp-SP is used, and assess any potential long-term benefits of the LAMP-based diagnostic intervention for malaria in pregnancy. Cost-effectiveness analyses are also required before implementation of LAMP in routine care.

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Contributors

DRP obtained funding for the trial. DRP, CK, RFG, and AGB wrote the trial protocol. DRP, DY, MT, CK, AGB, and BT managed the overall trial. FM, BS, JB-G, and ZM conducted field visits and managed trial sites. DRP, MT, JB-G, HT, NS, BT, and DY accessed and verified the underlying data and did the statistical analyses. JB-G did the laboratory analyses. All authors contributed to writing and reviewing the manuscript. The principal investigators (MT, BT, DY, and DRP) and LAMPREG trial team had full access to all data and had responsibility for the final decision to submit for publication.

Declaration of interests

DRP has received unrestricted educational and travel funds from diagnostic companies (Becton Dickinson, Biomerieux, New England Biolabs, and Abbott) and is an advisor to a University of Calgary start-up company, Illucidx. All other authors declare no competing interests.

Data sharing

De-identified participant data will be made available with a data sharing agreement after publication. The data sharing agreement would require approval of both institutions and investigators.

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References

1 WHO. World malaria report 2024: addressing inequity in the global malaria response. 2024. https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2024 (accessed June 30, 2025).
2 WHO. Global Technical Strategy for Malaria 2016–2030, 2021 update. 2021. https://www.who.int/publications/i/item/9789240031357 (accessed June 30, 2025).
3 Ethiopian Ministry of Health. Health and Health-related Indicators 2015 EFY (2022/2023 GC). Addis Ababa: Ethiopian Ministry of Health, 2023.
4 Desai M, ter Kuile FO, Nosten F, et al. Epidemiology and burden of malaria in pregnancy. *Lancet Infect Dis* 2007; **7:** 93–104.
5 Rogerson SJ, Hviid L, Duffy PE, Leke RF, Taylor DW. Malaria in pregnancy: pathogenesis and immunity. *Lancet Infect Dis* 2007; **7:** 105–17.
6 Newman RD, Hailemariam A, Jimma D, et al. Burden of malaria during pregnancy in areas of stable and unstable transmission in Ethiopia during a nonepidemic year. *J Infect Dis* 2003; **187:** 1765–72.
7 WHO. WHO guidelines for malaria. Nov 30 2024. https://iris.who.int/bitstream/handle/10665/379635/B09146-eng.pdf (accessed June 30, 2025).
8 Dickson KS, Okyere J, Ahinkorah BO, et al. Skilled antenatal care services utilisation in sub-Saharan Africa: a pooled analysis of demographic and health surveys from 32 countries. *BMC Pregnancy Childbirth* 2022; **22:** 831.
9 Kitojo C, Gutman JR, Chacky F, et al. Estimating malaria burden among pregnant women using data from antenatal care centres in Tanzania: a population-based study. *Lancet Glob Health* 2019; **7:** e1695–705.
10 Mbacham HF, Mosume DM, Apinjoh TO, et al. Sub-microscopic *Plasmodium falciparum* parasitaemia, dihydropteroate synthase (*dhps*) resistance mutations to sulfadoxine-pyrimethamine, transmission intensity and risk of malaria infection in pregnancy in Mount Cameroon Region. *Malar J* 2023; **22:** 73.
11 Hounkonnou CPA, Briand V, Fievet N, et al. Dynamics of submicroscopic *Plasmodium falciparum* infections throughout pregnancy: a preconception cohort study in Benin. *Clin Infect Dis* 2020; **71:** 166–74.
12 Okell LC, Griffin JT, Roper C. Mapping sulphadoxine-pyrimethamine-resistant *Plasmodium falciparum* malaria in infected humans and in parasite populations in Africa. *Sci Rep* 2017; **7:** 7389.
13 Ethiopian Public Health Institute. June 2019. https://ephi.gov.et/wp-content/uploads/2014/04/Evidence-brief-on-ITTp-22-2.pdf (accessed July 31, 2025).
14 Bain LE, Aboagye RG, Dowou RK, Kongnyuy EJ, Memiah P, Amu H. Prevalence and determinants of maternal healthcare utilisation among young women in sub-Saharan Africa: cross-sectional analyses of demographic and health survey data. *BMC Public Health* 2022; **22:** 647.
15 Endalamaw A, Engeda EH, Ekubagewargies DT, Belay GM, Tefera MA. Low birth weight and its associated factors in Ethiopia: a systematic review and meta-analysis. *Ital J Pediatr* 2018; **44:** 141.
16 van Eijk AM, Stepniewska K, Hill J, et al. Prevalence of and risk factors for microscopic and submicroscopic malaria infections in pregnancy: a systematic review and meta-analysis. *Lancet Glob Health* 2023; **11:** e1061–74.
17 Yon JLT, Htet NH, Naing C, Tung WS, Aung HH, Mak JW. Loop-mediated isothermal amplification (LAMP) test in the detection of uncomplicated malaria in pregnancy: a meta-analysis of diagnostic accuracy. *Malar J* 2022; **21:** 391.
18 Briggs J, Ategeka J, Kajubi R, et al. Impact of microscopic and submicroscopic parasitemia during pregnancy on placental malaria in a high-transmission setting in Uganda. *J Infect Dis* 2019; **220:** 457–66.
19 Chen I, Clarke SE, Gosling R, et al. "Asymptomatic" malaria: a chronic and debilitating infection that should be treated. *PLoS Med* 2016; **13:** e1001942.
20 Bousema T, Okell L, Felger I, Drakeley C. Asymptomatic malaria infections: detectability, transmissibility and public health relevance. *Nat Rev Microbiol* 2014; **12:** 833–40.
21 Walther B, Miles DJ, Crozier S, et al. Placental malaria is associated with reduced early life weight development of affected children independent of low birth weight. *Malar J* 2010; **9:** 16.
22 Lindblade KA, Steinhardt L, Samuels A, Kachur SP, Slutsker L. The silent threat: asymptomatic parasitemia and malaria transmission. *Expert Rev Anti Infect Ther* 2013; **11:** 623–39.
23 Gebresenbet RF, Kamaliddin C, Bekele ZM, et al. Active case detection and treatment of malaria in pregnancy using LAMP technology (LAMPREG): a pragmatic randomised diagnostic outcomes trial-study protocol. *BMJ Open* 2022; **12:** e058397.
24 WHO. Malaria parasite counting. Malaria microscopy standard operating procedure – MM-SOP-09. 2016. https://www.who.int/publications/i/item/HTM-GMP-MM-SOP-09 (accessed July 10, 2025).
25 Mohon AN, Getie S, Jahan N, Alam MS, Pillai DR. Ultrasensitive loop mediated isothermal amplification (US-LAMP) to detect malaria for elimination. *Malar J* 2019; **18:** 350.
26 Okwaraji YB, Krasevec J, Bradley E, et al. National, regional, and global estimates of low birthweight in 2020, with trends from 2000: a systematic analysis. *Lancet* 2024; **403:** 1071–80.
27 Briand V, Saal J, Ghafari C, Huynh BT, et al. Fetal growth restriction is associated with malaria in pregnancy: a prospective longitudinal study in Benin. *J Infect Dis* 2016; **214:** 417–25.
28 Garrison A, Boivin MJ, Fievet N, et al. The effects of malaria in pregnancy on neurocognitive development in children at 1 and 6 years of age in Benin: a prospective mother-child cohort. *Clin Infect Dis* 2022; **74:** 766–75.
29 Ethiopian Public Health Institute (EPHI); Ministry of Health, Ethiopia; Countdown to 2030. Health Sector Transformation plan-I (HSTP-I): endline review study. 2022. https://ephi.gov.et/wp-content/uploads/2023/01/2.-Ethiopia-Health-Sector-Transformation-Plan-I-2015-2020-Endline-Review.pdf (accessed July 11, 2025).